

CHAPTER 8: PERIPHERALS & REAL-TIME SYSTEMS

PERIPHERALS & REAL-TIME COMPUTING

This chapter examines two related topics:

1. **Peripherals:** Devices used for storage and interaction, focusing on their physical characteristics.
2. **Real-Time Systems:** Computing systems where the *timing* of results is just as important as the correctness of the results.

STORAGE DEVICES OVERVIEW

- **Magnetic Storage:** Uses magnetic patterns on a surface (e.g., **Hard Disk Drive (HDD)**, Magnetic Tape).
- **Optical Storage:** Uses lasers to read/write pits and lands on a reflective disc (e.g., **CD-ROM**, DVD, Blu-ray).
- **Solid-State Storage:** Uses non-volatile flash memory (e.g., **Solid-State Drive (SSD)**, USB drive).

HARD DISK LAYOUT (DIAGRAM)

An HDD is organized physically as follows:

- **Platters:** The physical disks that rotate, coated in magnetic material.
- **Tracks:** Concentric circles on a platter where data is stored.
- **Sectors:** Pie-shaped wedges of a platter.
- **Block (or Cluster):** The smallest unit of storage, formed by the intersection of a track and a sector.
- **Cylinder:** A set of all tracks that are at the same distance from the center, spanning all platters. (The read/write heads move as one unit).

CD-ROM LAYOUT

- Unlike an HDD, a CD-ROM does **not** use concentric tracks.

- It stores data in a **single, long, spiral track** that starts from the center and unwinds to the outer edge.
- Data is stored as microscopic "pits" (burned by a laser) and "lands" (unburned areas). A "1" or "0" is registered by the *change* from a pit to a land, or vice-versa.

CLASSIFICATION OF PERIPHERALS

Peripherals are devices external to the CPU and main memory. They are broadly classified by their function:

- **Input Devices:** Send data *to* the computer (e.g., keyboard, mouse, scanner, microphone).
- **Output Devices:** Receive data *from* the computer (e.g., monitor, printer, speakers).
- **Storage Devices:** Can both read and write data; used for non-volatile storage (e.g., HDD, SSD, USB drive).
- **Communication Devices:** Transmit and receive data with other computers (e.g., Network Interface Card (NIC), Wi-Fi adapter).

HARD VS. SOFT REAL-TIME SYSTEMS

The key differentiator is the **consequence of missing a deadline**.

HARD REAL-TIME SYSTEM:

- **Definition:** Missing a deadline is considered a **total system failure**. The consequences are often catastrophic.
- **Constraints:** Timing constraints are absolute and *must* be met. The system must be predictable and deterministic.
- **Example: Air Traffic Control** (a late radar update is useless and dangerous), anti-lock braking systems (ABS) in a car, a heart pacemaker.

SOFT REAL-TIME SYSTEM:

- **Definition:** Missing a deadline is **not a failure**, but results in **degraded performance**. The system continues to function.
- **Constraints:** The system *tries* to meet deadlines, but occasional misses are acceptable.

- **Example: Video Streaming** (missing a deadline results in a dropped frame or a moment of buffering, not a crash), **ATM** (if a transaction takes 2 seconds instead of 1, it's annoying but not a catastrophe), video games.

EXAMPLES: ATM, AIR TRAFFIC CONTROL, STREAMING

- **Air Traffic Control: Hard Real-Time.** Data *must* be acquired, processed, and displayed within a strict time window to prevent accidents.
- **ATM (Automated Teller Machine): Soft Real-Time** (or sometimes "Firm" Real-Time). The system *should* respond quickly for good user experience, but a delay of a few seconds won't cause a disaster.
- **Video Streaming: Soft Real-Time.** The system must decode and display frames at a steady rate, but if it's late, it simply drops a frame and continues.